

Cairo University

Faculty of Computers & Artificial Intelligence

Supervised learning

Assignment one

PCA and Feature Selection with Naive Bayes

Comparative Analysis Report

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1. Introduction

Machine learning models often suffer from high-dimensional datasets that contain redundant or irrelevant features. These unnecessary features may reduce model performance, increase computational cost, and introduce noise into the training process. Therefore, dimensionality reduction techniques are commonly used to improve model efficiency and accuracy.

Two common approaches for reducing dimensionality are feature selection and feature extraction. Feature selection removes irrelevant or redundant features while keeping the original variables. Feature extraction transforms the original variables into a smaller set of new variables that preserve most of the information in the dataset.

In this project, we evaluate the impact of dimensionality reduction on classification performance using Naive Bayes classifiers. Two datasets were used: a Breast Cancer dataset containing numerical medical measurements and a Lung Cancer survey dataset containing categorical survey responses.

2. Datasets Description

2.1 Breast Cancer Dataset

Source: UCI Machine Learning Repository – Breast Cancer Coimbra Dataset.

Dataset Characteristics:

Number of samples: 116

Number of features: 9

Target variable: Classification (Cancer / No Cancer)

Features include:

Age, BMI, Glucose, Insulin, HOMA, Leptin, Adiponectin, Resistin, MCP-1

These features are continuous numerical variables, which makes them suitable for Gaussian-based probabilistic models.

Class Distribution:

0 – Healthy

1 – Breast Cancer

2.2 Lung Cancer Survey Dataset**Dataset Characteristics:**

Number of samples: 309

Number of features: 15

Target variable: LUNG_CANCER

Example Features:

Gender, Age, Smoking, Yellow Fingers, Anxiety, Peer Pressure, Chronic Disease, Fatigue, Wheezing, Alcohol Consumption, Coughing, Chest Pain

Most features are categorical or ordinal values, typically encoded as 1 or 2.

Class Distribution:

YES – Lung cancer detected

NO – No lung cancer

3. Methodology

3.1 Naive Bayes Classifier

Naive Bayes is a probabilistic classifier based on Bayes' Theorem and the assumption that features are conditionally independent given the class.

Two versions of Naive Bayes were implemented from scratch.

3.2 Gaussian Naive Bayes

Gaussian Naive Bayes assumes that numerical features follow a normal distribution.

For each class and feature, the model computes the mean and variance. The likelihood is calculated using the Gaussian distribution.

Log probabilities are used to avoid numerical underflow during probability multiplication.

3.3 Categorical Naive Bayes

Categorical Naive Bayes is used for discrete features. The algorithm calculates the probability of each feature value given a class and multiplies them to obtain the posterior probability.

A very small probability is used to handle unseen values and prevent zero probability errors.

4. Feature Selection

Feature selection was applied to remove features that contribute little to classification performance.

Benefits of feature selection:

- Reduce noise
- Improve model accuracy
- Decrease computational cost
- Improve interpretability

Unlike PCA, feature selection retains the original variables.

5. Principal Component Analysis (PCA)

PCA is a feature extraction technique that transforms the original dataset into a smaller set of new variables called principal components.

Steps used in the implementation:

Step 1: Data Standardization

The dataset is centered by subtracting the mean from each feature.

Step 2: Covariance Matrix Calculation

The covariance matrix describes how features vary together.

Step 3: Eigen Decomposition

Eigenvalues and eigenvectors are computed. Eigenvectors define directions of the new feature space while eigenvalues measure variance.

Step 4: Sorting Components

Components are sorted based on eigenvalues from highest to lowest.

Step 5: Dimensionality Reduction

The dataset is projected onto the top k principal components.

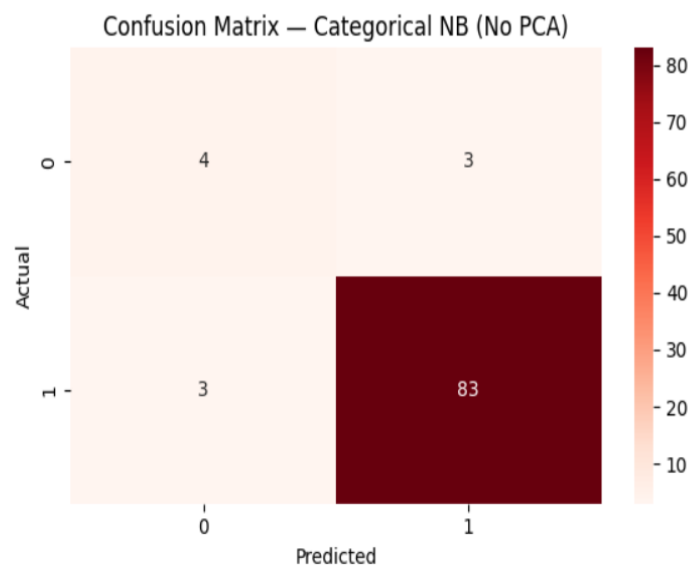
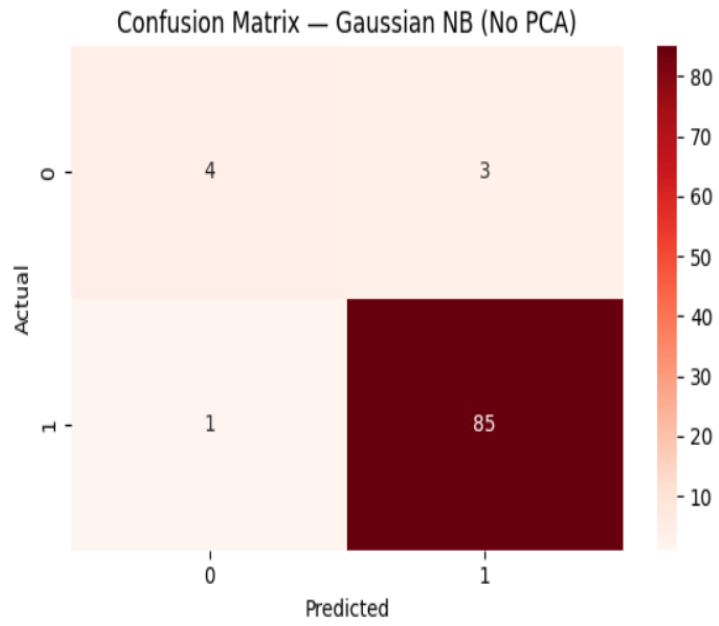
6. Experimental Results

Accuracy Comparison Table:

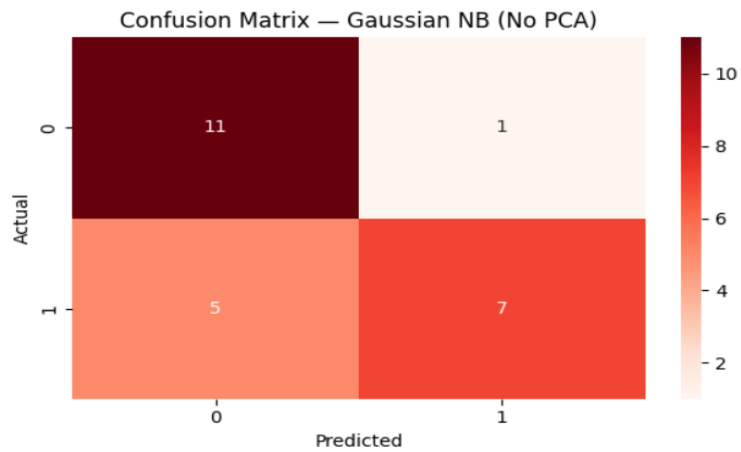
Model	Lung Cancer Dataset	Breast Cancer Dataset
Baseline (All Features + NB)	0.9570	.75
Feature Selection + NB	0.9355	0.7500
PCA + NB	0.7917	0.7917

Confusion matrices :

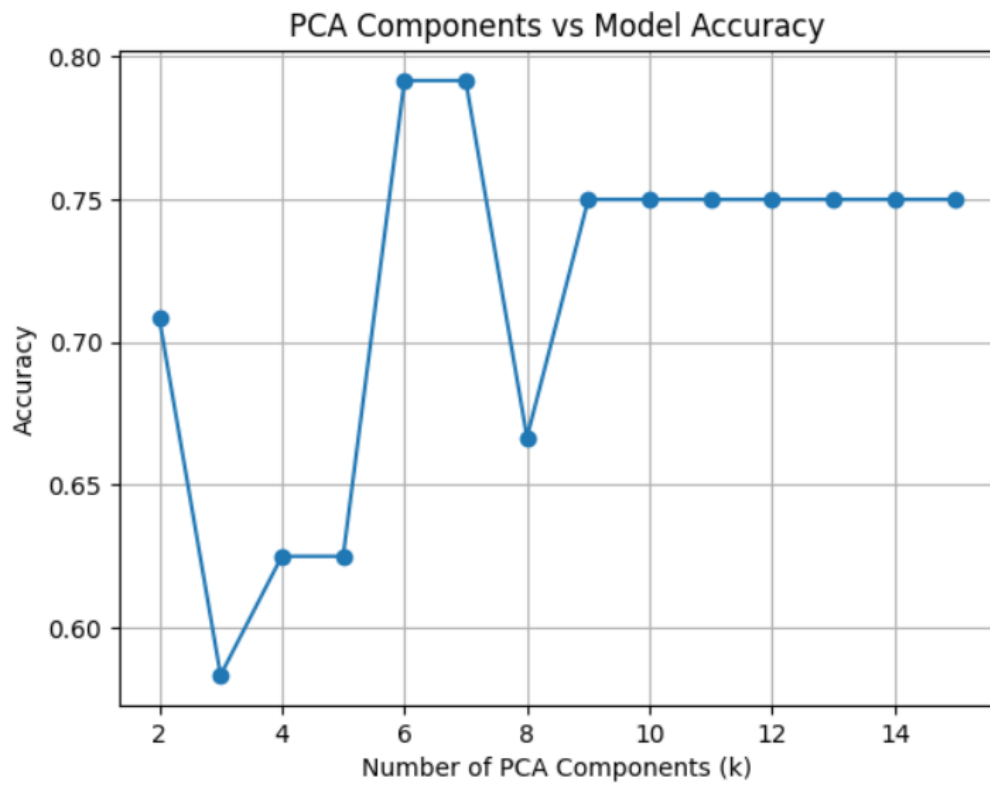
For dataset 1



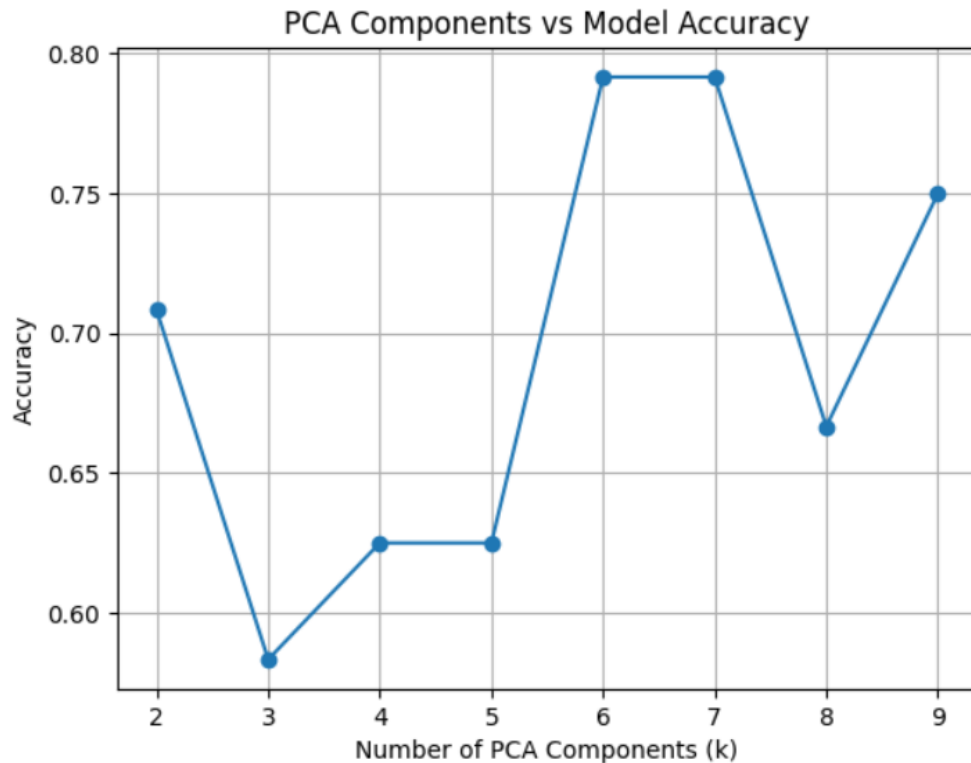
For dataset 2 :



A Scree Plot was also generated to show the variance explained by each PCA component.



Numerical dataset



7. Answers for Questions

- 1- Naive Bayes performs differently depending on data type. Gaussian Naive Bayes works well with numerical features, while Categorical Naive Bayes is more suitable for categorical datasets.
- 2- For the breast cancer dataset, PCA can improve performance by removing correlations and capturing the most informative variance.
- 3- For the lung cancer survey dataset, feature selection is usually more effective because PCA converts categorical variables into continuous components, which may distort relationships.
- 4- Feature selection preserves interpretability, while PCA reduces dimensionality more aggressively but produces less interpretable features.
- 5- PCA is generally not ideal for categorical data because it relies on variance and covariance, which are meaningful only for numerical variables.